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The effect of solution and gel forms of sodium hypochlorite on postoperative pain: a randomized clinical trial

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Introduction

Postoperative pain is a widespread complication after root canal treatments, which is undesirable for patients and physicians.[1] The incidence of postoperative pain stated in cases is 39% after endodontic treatment and in the first 24 hours, this rate can even be up to 65% and above.[2] Many mechanical, chemical and microbiological factors play a role in the occurrence of postoperative pain.[3] It especially appears as a result of extrusion of noninfected debris and solutions into periradicular tissues.[4] Many studies about this subject exists in the literature.[1] These studies have focused on different irrigation solutions and activation systems in addition to the mechanical preparation procedures of the root canal.[5,6]

Sodium hypochlorite (NaOCI) is the most widely used irrigation solution during root canal treatment.[3] In addition to its advantages such as antimicrobial activity and organic tissue solvency, however, it also has cytotoxic effects on periradicular tissues. When it is extruded from apical to periradicular tissues during root canal treatment, it damages endothelial cells and fibroblasts, facial nerve palsy, allergic reaction and necrosis may develop.[7] In a retrospective study conducted by members of the American Association of Endodontists, 42% of clinicians reported that using NaOCI causes postoperative pain or serious complications at least once a year.[8] That's why, researchers are in search of a more biocompatible irrigant.

It has been suggested currently that the use of the gel form of NaOCI is a potential option.[9,10] Studies have reported that the solution and gel forms of NaOCI have a similar effect on dentin. In the study of Zand, et al.[10] (2010) evaluating the smear layer removal activity of the solution and gel forms of NaOCI, and the study of Garcia, et al.[9] (2013) evaluating the effects of both forms on dentin microhardness, it has been reported that they have similar effects.

The effect of the gel form of NaOCI on postoperative pain has not been studied; therefore, this study compared the effects of using gel and solution forms of 5.25% NaOCI in mandibular molar teeth with symptomatic irreversible pulpitis on postoperative pain. The null hypothesis was that there would be no difference between the gel and solution forms of NaOCI.

Methodology

Study design, setting and sampling

Ethical approval for the study was obtained from the Institutional Review Board and the Ethics Committee of the University. In this clinical trial, Consolidated Standards of Reporting Trials guidelines were followed (Figure 1) and the study protocol was registered on www.clinicaltrial.gov (Identifier: NCT04190355). Participation in the study was voluntary. All patients signed an informed consent form after aims, procedures, benefits and potential risks of the study were explained. The study was conducted by 4 postgraduate students with the same level of experience, trained in endodontic procedures (using rotary instruments, irrigation and canal filling). Power and Sample Size Calculation software version 3.1.2 was used to calculate sample size. With 95% confidence, 95.1% test power and d=0.1121 effect size, the total sample size was determined to be 114.

Eligibility criteria

The inclusion criteria were as follows:

Healthy individuals aged 18-45 without any systemic disease;

Mandibular molar teeth that were diagnosed with symptomatic irreversible pulpitis that showed prolonged response in the tooth even after the removal of the thermal and electric pulp test;

Teeth with all root canals inclined up to 25° according to Schneider[11] (1971) method, with two canals in the mesial root and a single canal in the distal root;

Patients with preoperative pain scores between moderate and severe (VAS, 4-10) according to the VAS scale.

The exclusion criteria were the following:

Patients having taken analgesics or anti-inflammatory drugs in the past 12 hours;

Pregnant and lactating patients;

Teeth with radiologically proven periapical lesion;

Teeth that are too damaged to apply rubber dam;

Teeth with resorption, radiological evidence of calcification or open apices;

Patients with traumatic malocclusion;

Patients without occlusal contact.

114 patients were randomized into 2 groups based on the irrigation type during root canal preparation, using a computer program (available at

CONSORT 2010 checklist of information to include when reporting a randomised trial*

Section/Topic	Item No	Checklist item	Reported on page No
Title and abstract			
	1a	Identification as a randomised trial in the title	Title page
	1b	Structured summary of trial design, methods, results, and conclusions (for specific guidance see CONSORT for abstracts)	1
Introduction			
Background and objectives	2a	Scientific background and explanation of rationale	2
	2b	Specific objectives or hypotheses	2
Methods			
Trial design	3a	Description of trial design (such as parallel, factorial) including allocation ratio	3
	3b	Important changes to methods after trial commencement (such as eligibility criteria), with reasons	3
Participants	4a	Eligibility criteria for participants	3
	4b	Settings and locations where the data were collected	3
Interventions	5	The interventions for each group with sufficient details to allow replication, including how and when they were actually administered	3-5
Outcomes	6a	Completely defined pre-specified primary and secondary outcome measures, including how and when they were assessed	5
	6b	Any changes to trial outcomes after the trial commenced, with reasons	5
Sample size	7a	How sample size was determined	3
	7b	When applicable, explanation of any interim analyses and stopping guideline	-
Randomisation:			
Sequence generation	8a	Method used to generate the random allocation sequence	3-4
	8b	Type of randomisation; details of any restriction (such as blocking and block size)	3-4
Allocation concealment mechanism	9	Mechanism used to implement the random allocation sequence (such as sequentially numbered containers), describe in g any steps taken to conceal the sequence until interventions were assigned	4
Implementation	10	Who generated the random allocation sequence, who enrolled participants, and who assigned participants to interventions	4
Blinding	11a	If done, who was blinded after assignment to interventions (for example, participants, care providers, those assessing outcomes) and how	3
	11b	If relevant, description of the similarity of interventions	4
Statistical methods	12a	Statistical methods used to compare groups for primary and secondary outcomes	5
	12b	Methods for additional analyses, such as subgroup analyses and adjusted analyses	5
Results			
Participant flow (a diagram is strongly recommended)	13a	For each group, the numbers of participants who were randomly assigned, received intended treatment, and were analysed for the primary outcome	5
	13b	For each group, losses and exclusions after randomisation, together with reasons	5
Recruitment	14a	Dates defining the periods of recruitment and follow-up	5
	14b	Why the trial ended or was stopped	-
Baseline data	15	A table showing baseline demographic and clinical characteristics for each group	11
Numbers analysed	16	For each group, number of participants (denominator) included in each analysis and whether the analysis was by original assigned groups	12
Outcomes and estimation	17a	For each primary and secondary outcome, results for each group, and the estimated effect size and its precision (such as 95% confidence interval)	3-5
	17b	For binary outcomes, presentation of both absolute and relative effect sizes is recommended	-
Ancillary analyses	18	Results of any other analyses performed, including subgroup analyses and adjusted analyses, distinguishing pre-specified from exploratory	5
Harms	19	All important harms or unintended effects in each group (for specific guidance see CONSORT for harms42)	-
Discussion			
Limitations	20	Trial limitations, addressing sources of potential bias, imprecision, and, if relevant, multiplicity of analyses	8
Generalisability	21	Generalisability (external validity, applicability) of the trial findings	8
Interpretation	22	Interpretation consistent with results, balancing benefits and harms, and considering other relevant evidence	8
Other information			
Registration	23	Registration number and name of trial registry	-
Protocol	24	Where the full trial protocol can be accessed, if available	-
Funding	25	Sources of funding and other support (such as supply of drugs), role of funders	-

*We strongly recommend reading this statement in conjunction with the CONSORT 2010 Explanation and Elaboration for important clarifications on all the items. If relevant, we also recommend reading CONSORT extensions for cluster randomised trials, non-inferiority and equivalence trials, non-pharmacological treatments, herbal interventions, and pragmatic trials. Additional extensions are forthcoming: for those and for up to date references relevant to this checklist, see www.consort-statement.org.

Figure 1- The Consolidated Standards of Reporting Trials checklist

www.randomizer.org). Each group was randomly and equally divided. The allocation ratio was 1:1. In the study, patients were not informed about the division and they were blinded. However, the clinicians could not be blinded due to the nature of the study.

Treatment procedure

Root canal treatments of all patients were performed in a single session. Dental anesthesia was achieved using local anesthetic solution (Ultracain DS Fort, Hoechst-Marian Roussel, Frankfurt, Germany) containing 4% articaine and 1:200000 epinephrine for inferior alveolar nerve blockade. After rubber dam isolation, endodontic access cavity was prepared using high-speed burs (Dentsply Maillefer, Ballaigues, Switzerland). The working length (WL) was determined using apex locator (Propex Pixi, Dentsply Maillefer) and confirmed to be 0.5-1 mm shorter than the "radiographic apex" by periapical radiographs. The root canals were mechanically prepared using ProTaper Next (Dentsply, Maillefer, Ballaigues, Switzerland) up to X3. All ProTaper Next files were used with an endodontic engine (X-Smart, Dentsply Sirona) at the torque and speed values recommended by the manufacturer. After reaching the WL with size-15 K-type hand file, shaping was continued with brushing motion until the canal length was achieved with X1, X2 and X3 files, respectively. The files were withdrawn at the point where resistance was met before torsional overload occurred and the work continued after the apical opening was checked with size-10 K-type hand file. All patients were divided into two groups based on irrigant used during root canal preparation (n = 57): Group 1, 5.25% NaOCI solution (Imicrly, Konya, Turkey), Group 2, 5.25% Chloraxid gel.

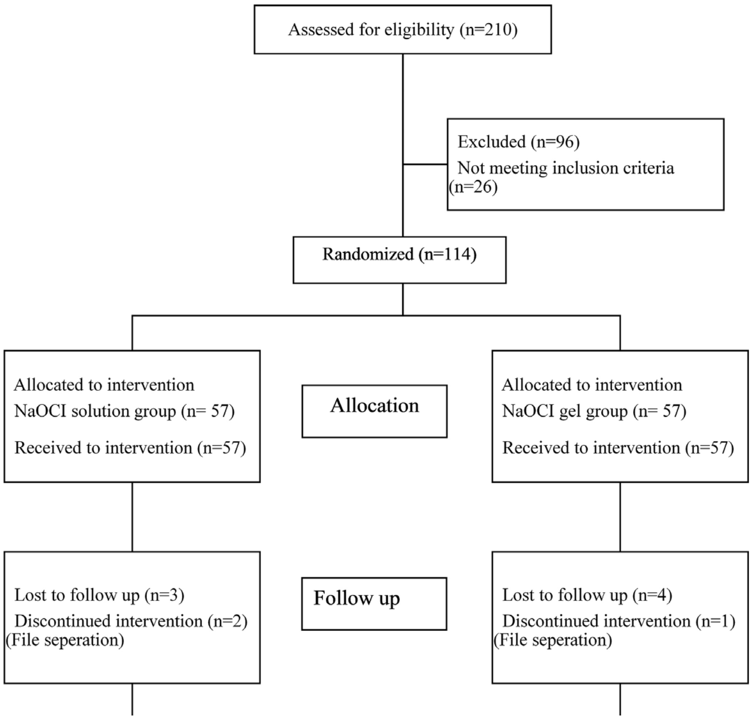
Group 1 (NaOCI solution): The root canals were irrigated with 5 mL of 5.25% NaOCI.

Group 2 (NaOCI gel): Gel form of 5.25% NaOCI (Chloraxid gel, Cercamed, Stalowo Wolo, Poland) was used. After covering the root canal files with NaOCI gel, they were placed in the root canal. During instrumentation canals were irrigated with 5 mL saline.

The irrigation procedure in all groups was performed with a NaviTip irrigation needle (30-G; Ultradent Products Inc, South Jordan, UT), placed 2 mm short of the working length. Once the shaping of the root canals was completed, according to the final irrigation procedure, all canals were irrigated with 5 mL of 17% EDTA solution (Imicryl, Konya, Turkey), 5 mL of 5.25% NaOCI solution and 5 mL of saline, respectively. Root canals were dried with absorbent paper points (Denstply Maillefer, Baillagues, Switzerland). All groups were filled with gutta-percha (Denstply Maillefer, Baillagues, Switzerland) and AH Plus Sealer (Denstply Maillefer, Baillagues, Switzerland) root canal paste in the same session using the single-cone technique. After the quality of obturation was ensured with radiographs, coronal seal was provided with glass ionomer cement (Amalgomer, AHL, Kent, UK). The teeth was restored with composite resin (Filtek Z250, 3M ESPE, St. Paul, Minnesota, USA) and, then, occlusal contacts was checked and relieved where necessary. Each patient was prescribed 400 mg of Ibuprofen and was instructed to take it every 8 hours when felt too severe and extremely unbearable pain to perform his daily activities.

Postoperative pain assessment

Visual analogue scale (VAS) was used for postoperative pain assessment. After endodontic treatment, all patients were given a detailed form to record their postoperative pain levels at the 6th, 24th, 48th, 72nd hours, and 1 week later. VAS assessment in this form was explained to the patients in detail, and they were asked to mark on the form the pain they felt at the 6th, 24th, 48th, 72nd hours, and 1 week later. One week after the treatment, patients were called by phone. The pain scores that the patients marked on the pain assessment form were learned and recorded in the patient file. According to the values recorded on the VAS, the pain levels were classified as no pain (0), mild pain (1-3), moderate pain (4-6) and severe pain (7-10).


Assessed for eligibility (n=210)
Excluded (n=96)
Not meeting inclusion criteria (n=26)
Randomized (n=114)
Allocated to intervention
NaOCI solution group (n= 57)
Received to intervention (n=57)
Allocation
Allocated to intervention
NaOCI gel group (n= 57)
Received to intervention (n=57)
Lost to follow up (n=3)
Discontinued intervention (n=2)
(File seperation)
Follow up
Lost to follow up (n=4)
Discontinued intervention (n=1)
(File seperation)

Conclusion

Within the limitations of the study, using gel or solution forms of NaOCI during chemomechanical preparation of root canals, resulted in similar postoperative pain. For both formulations, pain level decreased over time.

Conflict of interest

The authors deny any conflicts of interest related to this study.

Authors' contributions

Özlek, Esin: Conceptualization (Lead); Formal analysis (Lead); Investigation (Equal); Methodology (Equal); Project administration (Lead); Writing-original draft (Lead). **GÜNDÜZ, Hüseyin:** Conceptualization (Supporting); Data curation (Equal); Investigation (Equal); Methodology (Equal). **Kadi, Gizem:** Data curation (Equal); Investigation (Equal). **Tasan, Ahmet:** Data curation (Equal). **Akkol, Elif:** Data curation (Supporting); Formal analysis (Supporting).

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